Table of Contents

Experiment 1A: Introduction to MATLAB 1
Step 3: Create variables and assign numerical values 1
Step 4: Perform arithmetic operations 1
Step 5: Evaluate built-in mathematical functions 1
Step 6: Define symbolic variables and expressions 2
Step 7: Perform symbolic computations 2
Step 8: Generate values for the independent variable 2
Step 9: Plot a standard mathematical function 2
Step 10: Add title, labels, legend, and grid 2
Step 11: Display numerical and symbolic results 3

Experiment 1A: Introduction to MATLAB

Aim: To perform basic arithmetic operations, symbolic computations, and plot a standard mathematical function.

```
clc;
clear;
close all;
```

Step 3: Create variables and assign numerical values

```
a = 15;
b = 4;
```

Step 4: Perform arithmetic operations

```
addition = a + b;
subtraction = a - b;
multiplication = a * b;
division = a / b;
modulus = mod(a,b);
powerValue = a^b;
```

Step 5: Evaluate built-in mathematical functions

```
squareRoot = sqrt(a);
exponential = exp(2);
logarithm = log(a); % Natural logarithm
sineValue = sin(pi/6);
cosineValue = cos(pi/3);
tangentValue = tan(pi/4);
```

Step 6: Define symbolic variables and expressions

```
syms x
f = x^2 + 5*x + 6;
```

Step 7: Perform symbolic computations

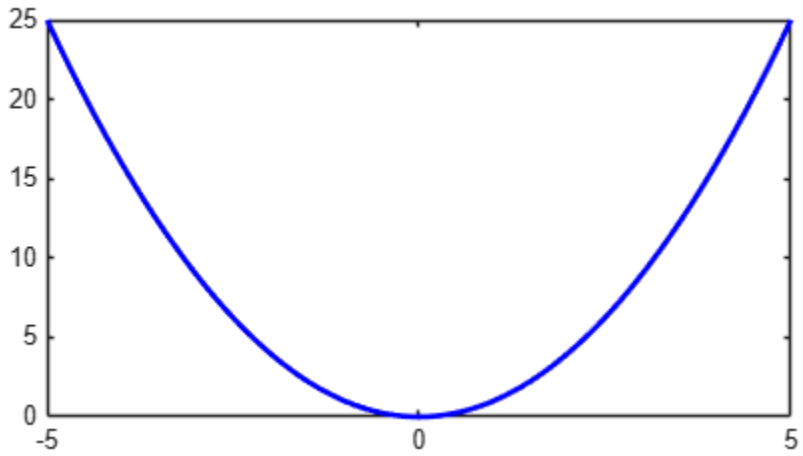
```
simplifiedExpr = simplify((x^2 - 1)/(x - 1));
expandedExpr = expand((x + 2)^3);
factoredExpr = factor(f);
```

Step 8: Generate values for the independent variable

```
xValues = -5:0.1:5;
```

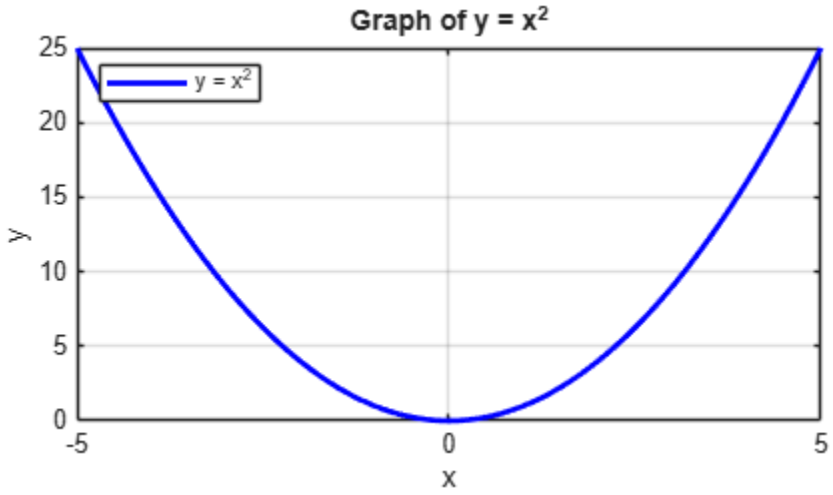
Step 9: Plot a standard mathematical function

```
yValues = xValues.^2;
figure;
plot(xValues, yValues, 'b', 'LineWidth', 2);
```



Step 10: Add title, labels, legend, and grid

```
title('Graph of y = x^2');
xlabel('x');
ylabel('y');
legend('y = x^2', 'Location', 'northwest');
grid on;
```



Step 11: Display numerical and symbolic results

```
disp('========================================');
disp(' Numerical Computations');
disp('========================================');

fprintf('Addition = %.2f\n', addition);
fprintf('Subtraction = %.2f\n', subtraction);
fprintf('Multiplication = %.2f\n', multiplication);
fprintf('Division = %.2f\n', division);
fprintf('Modulus = %.2f\n', modulus);
fprintf('Exponentiation (a^b) = %.2f\n', powerValue);

fprintf('\nSquare Root of a = %.4f\n', squareRoot);
fprintf('Exponential exp(2) = %.4f\n', exponential);
fprintf('Natural Log of a = %.4f\n', logarithm);
fprintf('sin(pi/6) = %.4f\n', sineValue);
fprintf('cos(pi/3) = %.4f\n', cosineValue);
fprintf('tan(pi/4) = %.4f\n', tangentValue);

disp(' ');
disp('========================================');
disp(' Symbolic Computations');
disp('========================================');

disp('Original Expression:');
disp(f);

disp('Simplified Expression:');
disp(simplifiedExpr);

disp('Expanded Expression:');
disp(expandedExpr);
```

```
disp('Factored Expression:');
disp(factoredExpr);

disp(' ');
disp('Program executed successfully.');
```

```
==========================================
 Numerical Computations
==========================================
Addition = 19.00
Subtraction = 11.00
Multiplication = 60.00
Division = 3.75
Modulus = 3.00
Exponentiation (a^b) = 50625.00

Square Root of a = 3.8730
Exponential exp(2) = 7.3891
Natural Log of a = 2.7081
sin(pi/6) = 0.5000
cos(pi/3) = 0.5000
tan(pi/4) = 1.0000

==========================================
 Symbolic Computations
==========================================
Original Expression:
x^2 + 5*x + 6

Simplified Expression:
x + 1

Expanded Expression:
x^3 + 6*x^2 + 12*x + 8

Factored Expression:
[x + 3, x + 2]


Program executed successfully.
```

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